Aerodynamic Interference in Confined Environments with Tilted Propellers: Wall Effect and Corner Effect

A. Garofano-Soldado, G. Heredia, A. Ollero

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***Abstract*—Inspection and maintenance tasks are increasingly being carried out by multi-rotor platforms in order to avoid certain risks being taken by humans. In this way, it is necessary to have a good knowledge of the aerodynamic effects that occur when tasks are performed in confined environments. In addition, the use of tilted rotors is becoming more and more widespread for tasks that require direct contact with a surface. In that sense, this paper presents several numerical simulations to analyse the aerodynamic performance of tilted rotors. In particular, the wall effect and the corner effect will be shown in detail. Two different configurations are considered in the corner effect: straight and curved corner. The influence of the corner and the wall has been studied for three tilt angles (θ = 20°, 30°, 40°) and various distances between the rotor and the wall. Flow field visualization is depicted to understand the physical behaviour of the airflow.**

***Index Terms*—UAV, aerodynamic performance, tilted rotor, CFD.**

I. Introduction

The use of unmanned aerial vehicles (UAV) has grown exponentially over the last few years [1]. Due to their good maneuverability and high reliability, these vehicles are used in a large number of applications, such as tracking, surveillance, inspection, maintenance, search and rescue, among others. Sometimes, aerial manipulation is used to carry out some of these tasks. This field of application is gaining a lot of interest from the scientific community [2], [3], [4]. The use of robotic arms minimizes the risk involved in some work at heights or in areas that are difficult to access, as well as reducing time and costs. In many of these tasks, UAV platforms have to operate close to surfaces such as walls, ground, ceiling, certain structures, or any kind of obstacle that may appear during the mission. Fig. 1 shows several UAVs perfoming inspections task (a) in a tunnel, (b) on a bridge [5], (b) and in a refiney [6]. In these cases, the aerodynamic interaction between the UAV and the environment causes the appearance of physical phenomena that must be compensated by the autonomous platform planner or controller in order to achieve an efficient and safe flight [7], [8].

In recent years, several studies have focused on the use of hexarotor configurations with tilted propellers because of the advantages over coplanar configurations for carrying out certain tasks such as direct contact inspection of infrastructures, inspection of bridges [9] or tunnels. Platforms with coplanar rotors are under actuated. In fact, they have to roll and pitch in order to move horizontally in a specific direction, whereas the use of inclined rotors allows the horizontal component of the force to be used to move without the need to tilt the platform [10]. In these cases, platforms have 6 degrees of freedom and are fully actuated [11]. In inspection and maintenance tasks where UAVs have to be very close to a surface or maintain direct contact with the surface to be inspected [5], the use of hexarotor platforms with tilted rotors are the best option to perform the task with enough safety and precision.

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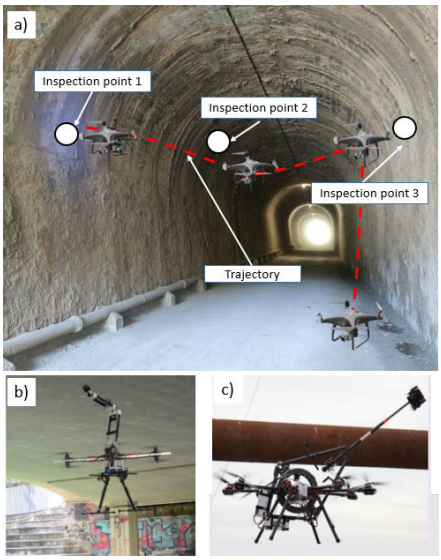


Fig. 1: Multi-rotors performing inspection and maintenance tasks: a) tunnel inspection, b) bridge inspection [5], c) refinery inspection [6].

In confined environments, the presence of surfaces or obstacles causes the rotor wake to be constrained and cannot be freely developed. Consequently, when the UAV approaches a surface, the rotor downwash modifies the flow field causing the thrust and torque forces of the rotors to change. Proximity effects can lead to destabilization of the UAV and must be compensated for to conduct safe operations. Although there are a large number of studies that analyse the aerodynamic effects of standard coplanar multi-rotor configurations when in contact with certain surfaces such as the ground [12], [13], [14] ceiling [8], [9], [15], wall or corner [16], [17] there are no studies that show the variation of thrust with distance to the ground or wall when using tilted rotors. In those cases, aerodynamic interference has a different behaviour to that presented in previous work and must be properly analysed to ensure the control stability of the multi-rotor. These aerodynamic effects can be examined by experimental studies, by performing tests in a wind tunnel, on a test bench

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specifically designed to study the aerodynamic influence of certain configurations [18], or by using Computational Fluid of Dynamics methods (CFD). Numerical simulations allow the flow field to be visualized and are able to provide a good understanding of the physical phenomena present in this type of analysis [19], [20], [21], [12].

The main contribution of this work is a detailed CFD analysis of the aerodynamic effects of tilted rotors in corner and wall environment. In addition, the flow field will be studied for different rotor inclinations and distances to the wall in order to understand the physical behaviour of the fluid when the wake of a tilted rotor interacts with a surface. Therefore, so as to achieve flight stability, the research of the corner effect and wall effect is considered to play an important role in the proximity effect of the rotorcraft.

The rest of the paper is organized as follows: in Section II the numerical methodology followed in CFD to study different aerodynamic effects is presented, in Section III straight corner effect in rotors with different inclinations is described and the variations of thrust ratio with angle are obtained. In Section IV, curved corner effect in rotors is analysed, and thrust ratio results with tilted rotors are presented. Section V analyses the wall effect with several tilt angles and shows the variations of thrust coefficient results of a tilt rotor. In addition, the flow fields generated by the influence of wall and corner are discussed in Section III, IV and V. Last, conclusions and future work are presented in Section VI.

II. NUMERICAL METHODOLOGY

The numerical simulations have been carried out using the commercial software ANSYS FLUENT. Computational Fluid Dynamics (CFD) allows the modelling of turbulent flows around the propeller in both free and confined environments. The governing equations of fluid mechanics are based on the application of the principle of conservation of mass, momentum and energy. In this analysis, these equations have been simplified by considering the incompressible flow as having a Mach number of less than 0.3. Under this consideration, the Navier Stokes equations can be written as:

$$\nabla \cdot \vec{v} = 0 \tag{1}$$

$$\rho \frac{\partial \vec{v}}{\partial t} + \rho(\vec{v} \cdot \nabla)\vec{v} = -\nabla p + \mu \nabla^2 \vec{v} + \rho \vec{g} \tag{2}$$

where ρ is the density, p is the static pressure, $\vec{v}$ is the flow velocity, ρg are the gravitational body forces, being g the gravitational constant and μ is dynamic viscosity. These equations are solved for time-averaged flows where the velocity and pressure components are decomposed into mean and fluctuating parts. Thus, the Navier Stokes equations can be transformed to the Reynolds-Averaged Navier Stokes (RANS) equations. In this formulation, the term $-\rho\overline{v_i' v_j'}$ appears, known as Reynolds stress tensor [22]. In order to close the system of equations, it is necessary to define a turbulent model. For this research, the realizable $k-\varepsilon$ turbulent model has been chosen. It is a two-equation model that provides results with high accuracy for flows involving rotation, adverse pressure gradients, separations, and recirculation. The Standard Wall Functions are used to model the near-wall treatment. The pressure-based steady solver has allowed to solve the proposed problem using a coupled algorithm. Unlike segregated algorithms where each variable is solved sequentially, the algorithm used allows the pressure-based continuity equation and momentum equations to be solved in a coupled way. As a result, the convergence rate is improved and the solution is reached more quickly. The spatial discretization scheme used to discretise the diffusive and convective term is Green Gauss Node-Based Gradient. This scheme is more accurate when unstructured meshes are used although the computation time is longer. The second order upwind scheme is chosen for the moment. First order upwind is used for Turbulent Kinetic Energy (k) and Turbulent Dissipation Rate (ε). Finally, the PRESTO scheme is recommended as the pressure interpolation scheme for rotating flows.

The Multiple Reference Frame (MRF) method, is adopted to model the behavior of the fluid around a propeller in a confined environment [12], [23]. MRF is a steady-state approximation that provides accurate results with low computational costs. In this method, the computational domain is split into two regions: a rotating domain where there is a propeller and a stationary domain. Both areas are separated by an interface in order to connect both domains. A constant rotational speed of 6300 revolutions per minute (rpm) is induced in the inner rotating domain. This speed has been set as it is a speed typically used in real world environment.

In this research, several aerodynamic effects have been analysed: straight corner effect, curve corner effect and wall effect. These effects are considered only for tilted rotors, as coplanar rotors have been studied in detail in previous works. In order to obtain CFD results, different computational domains and mesh generation must be carefully defined for each proposed configuration. The modification of any geometrical parameter requires a new computational domain, mesh and setup. A total of 75 different simulations have been carried out for this aerodynamic analysis. In all cases, a T-Motor propeller with a diameter of 13 inch and fixed pitch of 4.4 inch has been used to model a single rotor.

III. STRAIGHT CORNER EFFECT

A. Geometry and Boundary Conditions

The computational domain is tunnel shaped with a width and height of 20R, where R is the propeller radius. The stationary domain is divided in two bodies, one for defining the global outer domain and another to refine the mesh in the areas close to the rotating domain. This makes possible to ensure that the element size changes smoothly between the two domains to accurately capture the flow and can visualize flow field. The height of this body is 2R and the width is 4.2R. The rotating domain is defined as a cylinder with a diameter of 1.1D, where D is the propeller diameter, and a height of 0.5R where the rotor is enclosed. The size of computational domain is shown in Fig. 2. The distance

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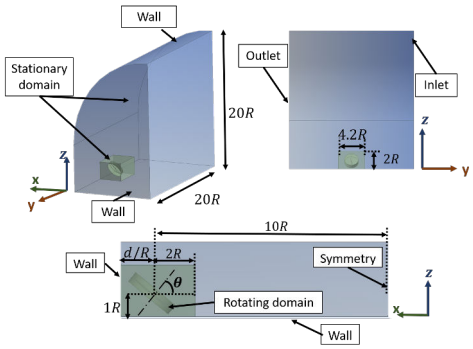


Fig. 2: Computational domain and boundary conditions in a straight corner.

between the rotor rotation center and wall is denoted by d. In this research, non-dimensional rotor-to-wall distance d/R is used:1.3, 1.4, 1.5, 1.8, 2.3, 2.8, 3.2, 3.8. The influence of this parameter has been studied for different rotor tilt angles, $\theta = 20°, 30°, 40°$. However, the distance between rotor rotation center and ground plane is fixed at 1R. The size of the inner stationary domain will change as the rotor move away from the wall.

The boundary conditions applied to analyse the straight corner effect are illustrated in Fig. 2. Pressure inlet is defined in frontal face with a gauge total pressure of 0Pa and pressure outlet is set in the opposite face with a gauge pressure of 0Pa. No-slip wall conditions are used in solid regions as ground plane, wall plane or propeller surface. Interface boundary conditions have to be defined between both domains to separate the rotational flow from the static flow. Finally, the symmetry condition is applied, and the stationary domain can be approximated to half a tunnel to reduce computational time.

B. Mesh independence study

The grid independence study must be performed to ensure that the solution is independent of the mesh definition [23]. For this purpose, several meshes have been defined and progressively improved with respect to the previously defined mesh. This process must be performed until the results do not change significantly or the improvement is not computationally profitable. Therefore, three simulations have been carried out considering different mesh densities.

A fully unstructured mesh was generated in both the stationary and rotating domain. Mesh creation must be done carefully as it plays an important role in the process of obtaining results. The convergence of the solution, the accuracy of the results and the computational time will be influenced by the quality of the computational grid. For the current study, element size in propeller and the rotating domain are the same for all configurations. In addition, the mesh has been created so that the smallest elements are in the vicinity of the propeller and the rotating region, increasing progressively towards the stationary domain.

Non-dimensional rotor-wall distance considered is 1.5R with a ground distance of 1R. In addition, the rotor tilt angle

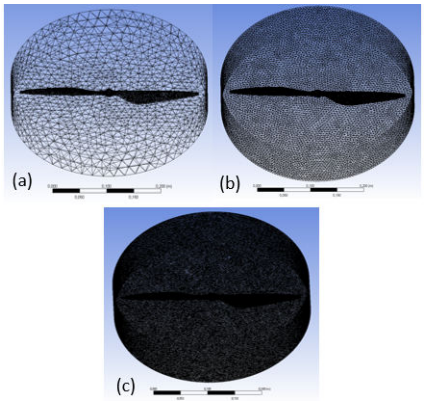


Fig. 3: Different mesh densities of the rotational domain: a) Mesh 1, b) Mesh 2, c) Mesh 3.

TABLE I: Grid independence study

Mesh densities		Rotating domain	Stationary domain
Mesh 1	Nodes	37.589	257.509
	Elements	207.571	1.498.357
Mesh 2	Nodes	213.687	297.766
	Elements	1.170.268	1.708.272
Mesh 3	Nodes	558.719	637.149
	Elements	3.015.656	3.538.461

is set as 20°. For this study, it has started with a low-density mesh (mesh 1) and then it has been increasing. The areas of interest are the rotating domain and the propeller surface. Fig. 3 shows different mesh densities of the rotational domain. Table 1 provides more information on the number of mesh elements and nodes for each of the three proposed meshes. The number of total elements changes significantly depending on the type of mesh made. In order to decide which mesh to use, the aerodynamic coefficients are compared. The thrust coefficient C_T and the torque coefficient C_Q can be defined as:

$$C_T = \frac{T}{\rho A \Omega^2 R^2}$$

$$C_Q = \frac{M}{\rho A \Omega^2 R^3}$$

where $\rho = 1.225 kg/m^3$ is the air density, R is the radius of the rotor, $A = \pi R$ is rotor disk area, Ω is rotational speed (rpm), T is thrust of the rotor (N) and Q is torque of the rotor (Nm). Table II shows the result of the thrust and torque coefficient for mesh 1, mesh 2 and mesh 3. It can be seen that the results of mesh 1 are lower than in the other cases, in particular, the thrust coefficient is 6.9% lower than the other meshes, and the torque coefficient is 15% higher. However, the results of meshes 2 and 3 have changed very little from each other. Therefore, mesh 2 will be chosen for this configuration as well as for the configurations presented in the following sections.

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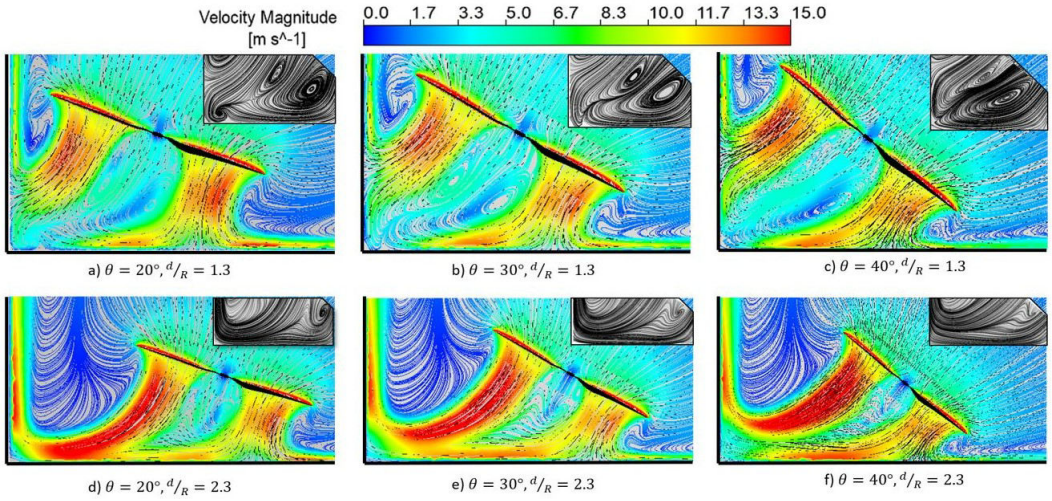


Fig. 4: Velocity magnitude contours to analyse the influence of the straight corner effect for different rotor inclinations and wall distances.

The total number of cells considered to study straight corner effect, curved corner effect and wall effect varies between 2.8 million for cases where the rotor is closer to the wall and 3.4 million for cases further away. In all simulations, the convergence criterion has been considered to be achieved when the value of the rotor thrust changes by 0.01% over 1000 iterations.

TABLE II: Thrust and torque coefficient

Mesh densities	C_T	C_Q
Mesh 1	0.00870	0.00110
Mesh 2	0.00934	0.00093
Mesh 3	0.00930	0.00093

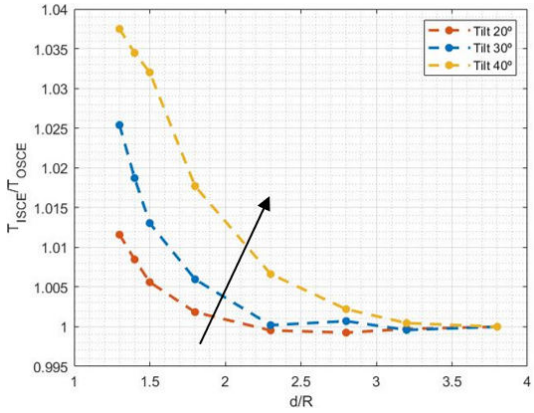


Fig. 5: Thrust incremented by the influence of a straight corner for different rotor inclinations and constant rotational speed. Data are connected with dashed lines.

C. CFD Results

Fig. 5 shows thrust ratio T_{ISCE}/T_{OSCE} versus the non-dimensional rotor-to-wall distance (d/R), where T_{ISCE} is thrust in straight corner effect and T_{OSCE} is thrust out straight corner effect at constant rotational speed for different tilt angles θ = 20°, 30° and 40°. These results are obtained for a constant rotor-to-ground distance. It can be seen that the thrust ratio is incremented when rotor approaches to the corner. As it increases tilt angle, the influence of the straight corner effect becomes greater. The simulations were carried out up to distance d/R = 3.8 since the results remain practically constant from this value onwards. It can be said the corner effect for values higher 3.8R is practically negligible for the angles analysed. In addition, distances of less than 1.3R from the wall were not considered because for some angles, the dimensions of the rotational domain did not allow this without interference between the rotating domain and the wall.

As illustrated in Fig. 5, for a tilt angle of 20°, small variations in thrust increment are observed between the case where the rotor is closest to the corner, for a distance of 1.3R, and the case furthest from the vertical wall, at a distance of 3.8R. In this case, the thrust is increased by 1.16% for the distance closest to the wall. As the rotor angle increases, the thrust ratio T_{ISCE}/T_{OSCE} is significantly higher in this confined environment. For an angle of 30°, the thrust increase is 2.5% and for an angle of 40°, this increment is almost 4%. Comparing this analysis with previous work on coplanar rotors, it can be seen that tilt rotor in confined environments behaves in a different way from what has been seen so far.

Fig. 4 illustrates the flow field in a straight corner environment generated for several rotor inclinations (20º, 30º and 40º respectively) and for wall distance of 1.3R and 2.3R. The flow streamlines and vectors are depicted in this figure to understand the physical phenomena that occur when fluid interacts with a surface. These results show the flow velocity magnitude in a plane orthogonal to the ground. It can be noted how the flow becomes more asymmetrical with respect to the

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axis of rotation as the angle of inclination of the rotor is raised and the formation of vortices due to the interference caused by the ground and the wall. The topology of the rotor wake changes significantly as the rotor moves away from the wall. In the upper right-hand corner of each picture of Fig. 4 the behavior of fluid when it is constrained can be viewed in more detail. For the distance 1.3R, two internal flow recirculation are created. Also, the size of these downward flows increases considerably when the tilt angle rises. This causes the thrust ratio to be higher for larger angles. However, for the distance 2.3R, the behaviour of fluid es completely different. When the rotor-wall distance is greater, the fluid is not so constrained and begins to expand. In these cases, smaller recirculation flows are generated in the regions close to the rotor hub causing lower thrust forces than for distance closer to the wall. The flow is dominated by higher velocity stream as shown in Fig. 4.

IV. CURVE CORNER EFFECT

A. Geometry and Boundary Conditions

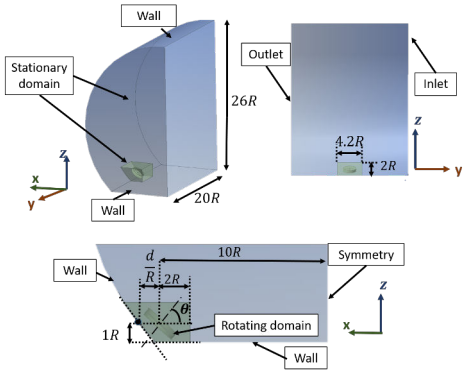


Fig. 6: Computational domain and boundary conditions in a curved corner.

The computational domain defined to study curved corner effect is illustrated in Fig. 6. As in the previous case, there are two domains: a stationary domain where there are two bodies and a fluid domain where a rotor is enclosed. In this region, a rotational speed is induced. The shape of this domain is different from the one defined in Section III. In this case, the tunnel has a circular shape until it interacts with the ground. Consequently, the corner generated is curve and the tangent to this curve at point P can be seen in Fig. 6. The external body of the stationary domain has a height of 26R and a width of 20R. Additionally, the distance between the rotor rotation center and the point P is denoted by d. The internal body of the stationary domain has a height of 2R and a width of 4.2R. In this study, the dimensionless distance d/R is used: 1.3, 1.4, 1.5, 1.8, 2, 2.24, 2.48, 2.67, 2.97. These distances have been set for different rotor inclinations (θ : 20°,30°,40°) in order to analyse the aerodynamic effects caused in each case. However, as in Section III, the rotor-to-ground distance is constant at 1R for all simulations. The size of the inner stationary domain will change as the rotor approaches the center of the tunnel.

Fig. 6 shows the boundary conditions applied to examine the curved corner effect. In frontal face, pressure inlet is set considering a gauge total pressure of 0Pa and in opposite face, pressure outlet is defined with a gauge pressure of 0Pa. No-slip wall condition is assumed in a solid boundary as ground, curved wall or propeller surface. Furthermore, the interface boundary condition allows to separate the rotational domain and stationary domain. Ultimately, the symmetry condition is applied to decrease the size of the computational domain and thus reduce computational effort.

B. CFD Results

Several numerical simulations have been performed to research the curved corner effect. In Fig. 7 it is illustrated the variation of the thrust ratio T_{ICCE}/T_{OCCE} versus dimensionless rotor-to-wall distance d/R, where T_{ICCE} is the thrust in curved corner effect and T_{OCCE} is the thrust out curved corner effect for different rotor inclinations ($\theta = 20°, 30°$ and $40°$).

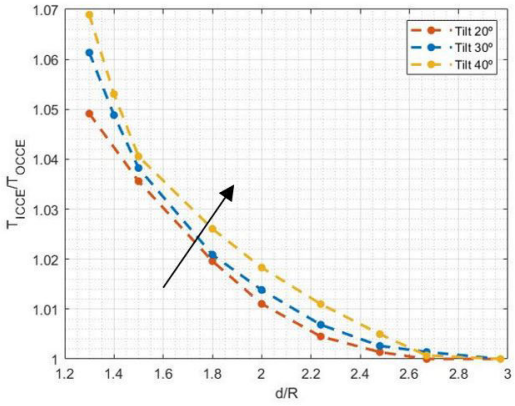


Fig. 7: Thrust incremented by the influence of a curved corner for different rotor inclinations and constant rotational speed. Data are connected with dashed lines.

The distance d/R cannot be lower than 1.3 to avoid that the rotational domain intersects with the wall. In addition, values greater 2.97 have not been simulated, since the change in the results is considered negligible below this value.

It can be observed that the trend of the results is very similar to those presented in the previous section where the thrust ratio increases as the rotor approaches the wall and as the rotor tilt angle is raised. However, in this case, the increment of thrust experienced by the rotor is higher than in the case of a straight corner. For instance, when the rotor is placed in the position closest to the wall (d/R=1.3), there is an increase of approximately 7%, 6.1% and 5% for angles of 40°,30° and 20°, respectively. In addition, as shown in Fig. 7, the thrust ratio changes smoothly as the angle of inclination becomes larger. For the case of the straight corner, the variation of thrust ratio with angle is more significant.

The velocity field generated by the interaction of the rotor flow with a curved corner is depicted in Fig. 8 for the d/R distances of 1.3R and 2.24R and for the tilt angles: 20º, 30º

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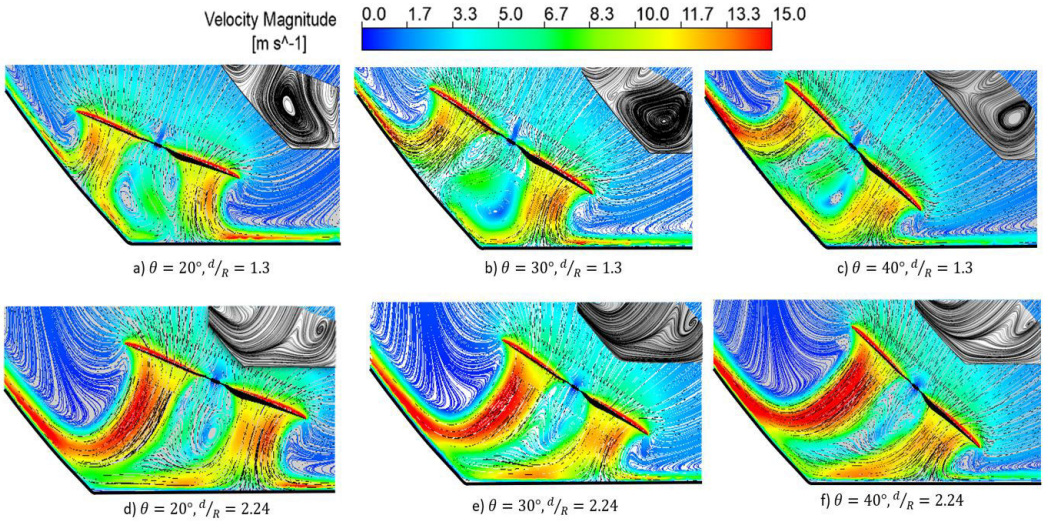


Fig. 8: Velocity magnitude contours to analyse the influence of the curved corner effect for different rotor inclinations and wall distances.

and 40º. As in the case of straight corner, the fluid is not symmetrical. The streamlines and vortices are represented to understand the behaviour of downwash wake when interacting with the curved corner. The fluid is compressed with the walls causing flow recirculation and the formation of large vortices when the distance to the wall is small. However, this bubble recirculation becomes smaller and smaller when the distance d/R is lower. In these cases, a vortex appears at the rotor hub. Comparing these results with those obtained in Section III, it can be seen that the vortices in the straight corner are made up mainly of two vortices, while the curved corner is notable for the formation of a larger vortex. The formation of vortices is directly reflected in how the thrust of tilt rotors changes as it approaches the corner. These effects must be taken into account, otherwise destabilization of the UAV will occur when performing tasks in a confined environment.

V. WALL EFFECT

A. Geometry and Boundary Conditions

The wall effect is examined in this section in order to analyse the interactions between a tilted rotor and a wall when the ground is far enough away. Fig. 9 shows the geometry of computational domain and boundary conditions considered to study the wall effect. In the same way, as in the previous sections, a stationary two-body domain and a circular rotating domain are created. The rotational speed of the induced fluid inside the rotational domain is 6300 rpm and remains constant throughout the analysis. As illustrated in Fig.9, the height of the external stationary domain is 27R and the width is 20R. The dimensions of the inner stationary domain will be larger than for the corner case as the whole expansion region of the fluid has to be captured. In addition, the volume of this domain will change as the rotor moves away from the wall. The non-dimensional distance d/R is used: 1.3, 1.4, 1.5, 1.8, 2.3, 2.8, 3.2, 3.8. The same distance as in section III have been used to analyse the influence of the ground on the wall effect. In this study, the ground distance is 15R. The rotor inclinations studied are 20°,30° and 40°.

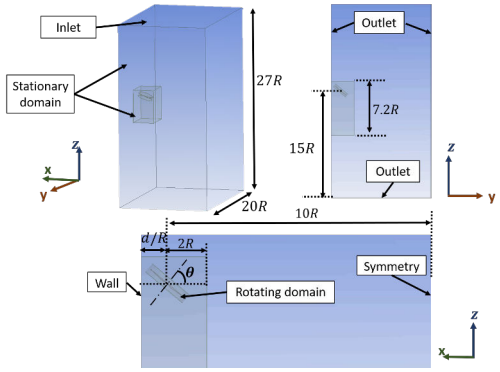


Fig. 9: Computational domain and boundary conditions in a wall.

Different boundary conditions are adopted in this scenario. The pressure inlet is set at the top of the domain. The front, opposite and bottom side of the domain are defined with the pressure outlet boundary condition. No-slip wall condition is applied on the propeller surface and vertical wall next to the rotor. Finally, the symmetry condition is assumed to split the domain and reduces computational efforts.

B. CFD Results

The aerodynamic performance of a tilt rotor when interacts with a wall are completely different from those obtained in a corner environment. In order to analyse the influence of the wall effect, the thrust ratio T_{IWE}/T_{OWE} versus non-dimensional distance d/R, where the ground is sufficiently far away is presented in Fig. 11. In this figure, T_{IWE} is the thrust in wall effect and T_{OWE} is the thrust out wall effect for different

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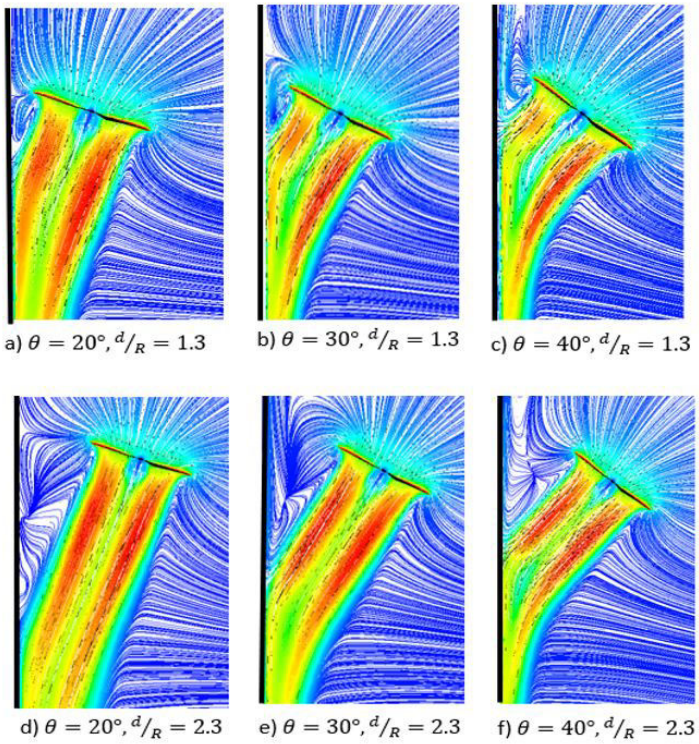


Fig. 10: Velocity magnitude contours to analyse the influence of the wall effect for different rotor inclinations and wall distances.

angles: θ = 20°, 30° and 40°. It can be seen how the wall

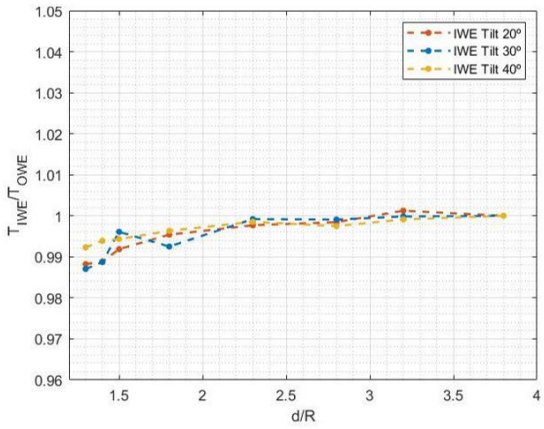


Fig. 11: Thrust incremented by the influence of a wall for different rotor inclinations and constant rotational speed. Data are connected with dashed lines.

effect reproduces the opposite behaviour to that presented in the straight corner effect in Fig. 5. Moreover, the variation of the thrust increment with the angle for the case considered in this section is very weak. In contrast to the corner effect, the trend of the results changes significantly. A decrease of thrust around 1% at a distance d/R= 1.3 for all inclinations is noted. In previous work, this effect was tested for standard coplanar rotor configuration, and the thrust results showed a similar trend to those presented in this study. Konus and Kanvas [24] discussed the rotor vortex wake in proximity to the wall. They found a 6% decrease as the three-bladed rotor approached the wall. The distances to the wall used by Konus and Kanvas were smaller than those used in this study because the dimensions of the computational domain do not allow the rotor to be brought closer to the wall. Han et al. [25] showed a decrease of 0.33% in C_T, Jardin et al [18] and Sánchez et al [17] considered the wall effect to be negligible for coplanar rotors. Fig. 10 shows the modified flow structure beneath the rotor for distance d/R = 1.3,2.3, as well as for angles 20°,30° and 40°. In this figure, the velocity magnitude is depicted by streamlines and vectors. In contrast to the corner environment, the rotor wake can be developed without significant flow recirculation regions. However, the flow cannot expand freely, and some source points are displayed in Fig. 10 as a consequence of the proximity of the wall. It is observed that for distance d/R = 1.3, the tip vortex become larger as the rotor inclination angle increases.

VI. CONCLUSION AND FUTURE WORKS

A large number of simulations has been carried out throughout this study in order to examine three aerodynamic effects: the straight corner effect, the curved corner effect, and the wall effect. In addition, different rotor inclinations and distances to the wall have been considered for each proposed scenario in static conditions. The following conclusions have been drawn from the results obtained using simulation techniques:

- In a straight corner and curved corner, thrust ratios T_{ISCE}/T_{OSCE} and T_{ICCE}/T_{OCCE} are increased as a tilted

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rotor approaches the wall. For the same inclination, it is observed that the curve corner causes a larger variation in thrust increment than in the case of a straight corner.

- As the tilt angle θ increases, thrust ratios for straight and curved corner become greater. Furthermore, it is noted that although T_{ICCE}/T_{OCCE} changes smoothly when the angle is modified, the increment of thrust experienced by the rotor is greater than when considering a straight corner.
- The thrust of a tilted rotor decreases when the ground effect is not taken into account and approaches the wall. This behaviour is contrary to that found for the other aerodynamic effects.
- The wall effect causes the rotor thrust variation with the angle to be slight. In that case, the tilt angle does not have a significant effect when the operating environment is in the vicinity of a wall. However, the use of inclined rotors makes the wall effect more prevalent than with coplanar rotors.

In future works, the corner effect will be studied for different ground distances in order to model the aerodynamic characteristics in these regions. In addition, control strategies will be used to compensate for these effects. The quantification of aerodynamic effects with tilted rotors must be taken into account for safer flights in confined environments and involving aerial manipulation tasks.

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